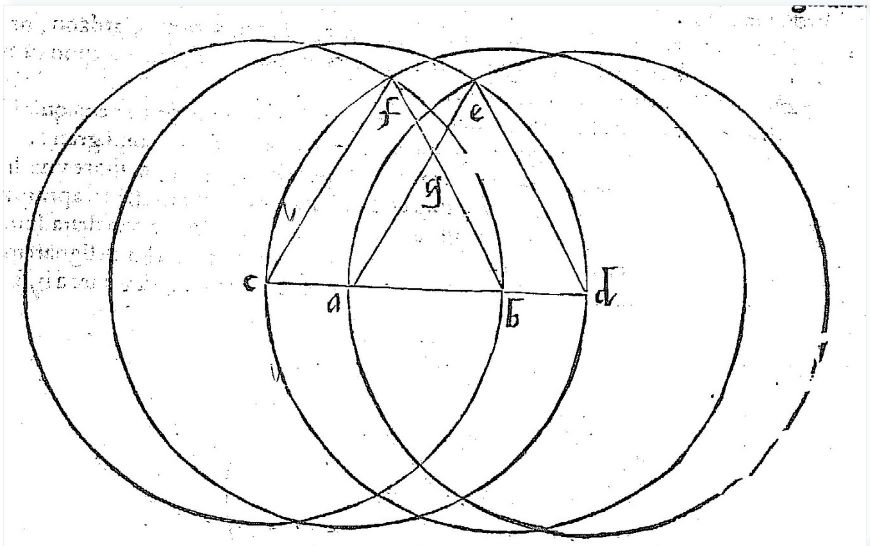
Quinta parte del *General Trattato* di Nicolò Tartaglia, Venezia 1560

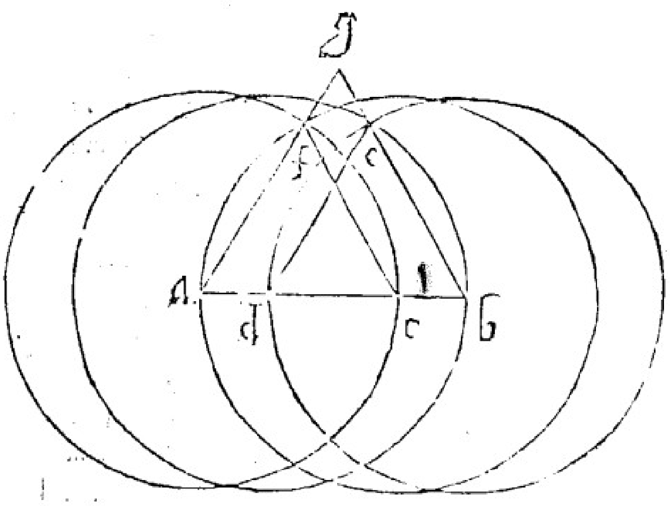
Libro Terzo, 63v-82r

We can construct an equilateral triangle with sides of any length determined by our opponent's compass opening. For example, suppose we want to construct an equilateral triangle with side length *ab*. I claim that we can construct such a triangle regardless of the fixed compass opening. If the compass opening were exactly *ab*, then by Euclid's Proposition I.1, we would follow that construction. However, if the fixed compass opening is greater than *ab*, then we draw two circles with centers at *a* and *b*, respectively.



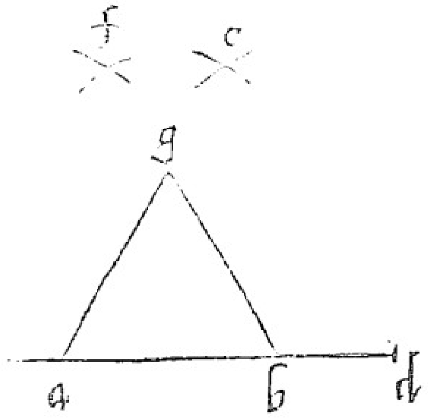
We extend the line in both directions until it intersects the circles at points c and d, and we draw triangle AED with side AD, which is equilateral **by Euclid's Proposition I.1.** In fact, you can draw a circle with center d using the compass opening. We can do the same by taking point c as the center and drawing the equilateral triangle cfb. The sides fb and ea intersect at point g, and the triangle g ab is equilateral because the two angles a and b are equal. In fact, they are the same angles as those of the equilateral triangles f b and a e d. Therefore, **by Euclid's Proposition I.32 or VI.4**, the triangle g ab is similar to the aforementioned equilateral triangles and is therefore equilateral.

But if the opening of the proposed compass is smaller than the given line AB, we shall proceed in the same manner as before. That is, with centers at the two endpoints A and B we shall describe two circles with the given opening, as shown below. These circles intersect the line AB at the two points C and D. Next, upon the two segments DB and AC we shall construct two equilateral triangles, according to the **first proposition of Euclid**, namely DBE and AFC. We shall then extend the two sides BE and AF until they meet at the



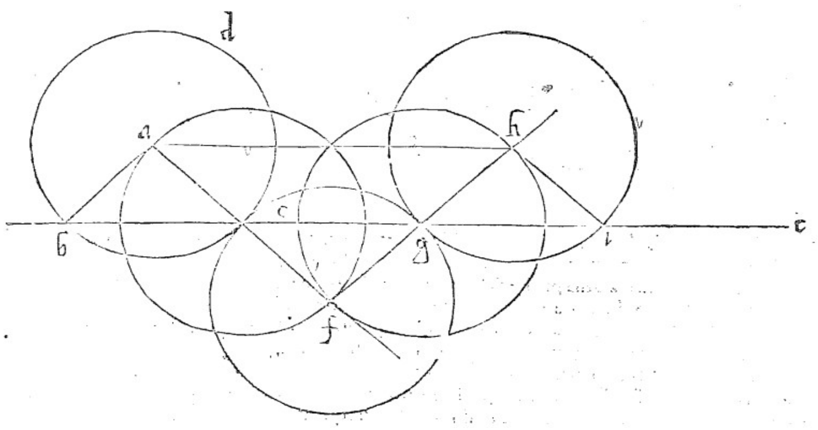
point G, thereby forming the triangle GAB, which, for the reasons given in the previous construction, will be similar to each of the other two equilateral triangles. Hence it too will be equilateral, which is the desired result.

The same construction applies when the opening of the compass is smaller than, or equal to, one half of the line AB.



It should be observed that the mere practical craftsman, who is not concerned with providing a geometrical demonstration of his construction, need not draw in full the two circles mentioned above. For his purpose it is sufficient, in the first construction, simply to mark the point C at a distance from the point B equal to the opening of his compass, and likewise the point D at the same distance from the point A. He then marks the intersection F from the points B and C with the compass, and similarly, from the points A and D, marks the intersection E. Having done this, he draws two faint construction lines, namely AE and BF, until they intersect at the point G. Then, from the point G, he draws the two lines GA and GB in ink, and thus obtains the desired result, namely the triangle GAB, which, provided he has worked carefully, he will find by direct experience to be correct.

If, however, one wishes to furnish the geometrical demonstration, one must proceed as I have shown above. [...]



Proposition 31 of the First Book of Euclid, not treated by Girolamo Cardano nor by his follower Ludovico Ferrari, and without which none of the seventeen problems that I proposed to them can be solved.

From a given point outside a given straight line, to draw a line parallel to it with any prescribed opening of the compass.

For example, let the point A be given outside the line BC. I say that from the point A one may draw a line parallel to the line BC with any opening of the compass. This opening is either greater than, less than, or equal to the distance (*differenza*) from the point A to the line BC.

First, let it be greater. With center at the given point A, let us describe the circle BED, which cuts the given line at the two points B and E. Let us draw the two lines AB and AE, thus forming the isosceles triangle ABE. We then extend the line AE indefinitely beyond E, and on it cut off the segment EF, equal to AE, that is, equal to the opening of our compass. Next, with center F, I describe the circle EG, which cuts the given line BC at the points E and G. Then we draw the line FG, extend it indefinitely beyond G, and in the same manner cut off the segment GH, equal to FG. With center H, I then describe the circle GI, which cuts the line BC (! *ab)* at the points G and I.

Finally, we draw the line HI, thus forming the triangle GHI. This triangle is equal and similar to the triangle BAE, since both are similar to the triangle EFG. For the angle at E of the one is equal to the angle at E of the other, **by Proposition 15 of Book I of Euclid**. Likewise, the angles at B and G are each equal to the said angle at E, **by Proposition 5 of Book** I, since the triangles are isosceles, having two equal sides. Moreover, **by Proposition 32 of Book I**, the angle at F is equal to the angle at A. Therefore, **by Proposition 4 of Book VI**, the triangles are equal and similar. By the same reasoning, the same follows for the triangle GHI.

Hence the triangle HGI is equal to the triangle ABE, and they stand upon the same straight line and on the same side. Therefore, **by Proposition 4 of Book I**, they lie between parallel lines. Consequently, if we draw the line AH from the point A to the point H, it will be parallel to the given line BC, which is the first desired result.